Agile Planning Document

CodeGrade: Smart Student Assignment & Code Reviewer

Group - 11

Batch - 8B

1. Project Vision:-

To design and build a smart, AI-assisted platform (CodeGrade) for teachers to manage coding assignments, and for students to write, test, and submit their code seamlessly. We plan to utilize a modern, decoupled architecture separating the frontend client from a secure backend REST API.

2. Sprint Breakdown:-

Sprint 1: Core Architecture & Authentication Goal:

Establish the decoupled foundation of the web application, database models, and secure user routing.

Backlog Items:

- **Initialize two separate projects:** an ASP.NET Core MVC frontend and an ASP.NET Core Web API backend.
- Configure Entity Framework Core and SQL Server contexts within the Web API.
- Implement **HttpClientFactory** in the MVC project to securely communicate with the API.
- Build Cookie-based Authentication flow (Login/Register).
- Enforce Role-Based Authorization (Teacher vs. Student routing).
- Design Landing Page with Bootstrap 5 UI/UX.

Sprint 2: Classroom & Assignment Management Goal:

Allow teachers to create digital learning spaces and populate them with assignments.

Backlog Items:

- Build Classrooms CRUD endpoints in the API and Teacher Dashboard UI in the MVC client.
- Implement Dynamic Join Code generator for Classrooms.
- Build Student Enrollment logic (Enrollments table) and tracking timestamps.
- Build Assignments CRUD endpoints.
- Implement dynamic **AllowedLanguage** filtering for assignments.

Sprint 3: The Coding Environment & Dashboards Goal:

Provide an integrated developer environment for students and data tracking for teachers.

Backlog Items:

- Integrate Microsoft Monaco Editor for in-browser coding.
- Implement dynamic syntax highlighting based on teacher's language rules.
- Integrate SweetAlert2 for polished, animated UI popups (e.g., "Code Submitted Successfully").
- Develop Teacher Submissions Dashboard with real-time JavaScript filtering.
- Engineer Matrix logic to cross-reference enrolled students against missing assignments.

Sprint 4: AI Automation & Feedback Loop Goal:

Integrate event-driven workflow automation to grade student code using the Google Gemini LLM.

Backlog Items:

- Set up a local n8n instance and configure a Webhook trigger node.
- Develop an **N8nService** in the MVC app to dispatch JSON payloads upon student code submission.
- Engineer a strict LLM system prompt for Google Gemini inside n8n to analyze code, check for bugs, and assign a grade.
- Create a dedicated **/grade** Web API receiver endpoint to accept the graded payload from n8n and update the database.
- Design a dedicated "Review Submission" page with a read-only Monaco Editor for both teachers and students to safely view the AI feedback and final score.